

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
		(iii)	Valid method for calculating the area (1) Correct value based on valid method (expect 6.0 ± 0.4 [J]) (1) Valid methods are: Counting squares expect 6 large squares, expect 24 of the $\frac{1}{4}$ squares and expect 600 tiny squares OR Trapezium (0,0), (0.3, 0), (0.3, 30), (0.2, 30) Note that this is equivalent to a triangle and a quadrilateral. Expect: $\frac{1}{2} \times (0.1 + 0.3) \times 30$ OR $\frac{1}{2} \times 0.2 \times 30 + 0.1 \times 30$ OR Trapezium (0,0), (0.3, 0), (0, 10), (0.2, 30) Note that this is equivalent to a triangle and a quadrilateral. Expect very similar numbers: $\frac{1}{2} \times (10 + 30) \times 0.3$ OR $\frac{1}{2} \times 20 \times 0.3 + 10 \times 0.3$ OR Triangle (0,0), (0.3, 0), (0.3, 40) Expect the numbers: $\frac{1}{2} \times 0.3 \times 40$ OR Triangle and trapezium method. Expect the numbers: $\frac{1}{2} \times 0.1 \times 20 + \frac{1}{2} \times (20 + 30) \times 0.2$ (= 1 J + 5 J) OR Anything else which is clearly correct award the marks e.g. triangle-trapezium-trapezium all with a base of 0.1 gives 6.1 [J] The final mark is for the correct answer 6.0 ± 0.4 [J]		2		2	2	2
		(iv)	Hysteresis (1) Difference between energy stored in the rubber band when it is stretched and [useful] energy recovered from it when it is unstretched. Or energy dissipated (or described) [lost] in one cycle of loading and unloading (1)	2			2		2
			Question 6 total	6	6	0	12	2	6